

AD2 | Rethinking Logistics

"Utilitati et Humanitati"

For Utility & Humanity

Rethinking Logistics examines how global logistics systems are being redesigned in response to disruption, urban growth, and sustainability demands.

The course focuses on transforming logistics hubs from static storage sites into dynamic, data-driven coordination centers. It explores hub redesign in terms of location strategy, automation, multimodal connectivity, and real-time digital integration.

Students study logistics as interconnected networks shaped by resilience, efficiency, and environmental constraints. Key themes include urban logistics pressures, micro-fulfillment models, AI-driven optimization, and circular supply chains.

By the end, learners understand how logistics infrastructure is evolving into adaptive systems that integrate technology, policy, and spatial planning strategies.

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This is a land-use structure map of Prague, Czech Republic.

It reveals a clear urban hierarchy, with a dense historic core surrounded by residential districts and extensive green landscapes.

From the center outward, the urban fabric gradually becomes less dense and more flexible in function, allowing for larger infrastructural and industrial developments. These peripheral areas offer larger available plots and fewer spatial restrictions, making them more suitable for large-scale logistics operations.

This area is Prague 9, a rapidly developing industrial district located on the edge of the city.

The area contains large open plots and former industrial land that are well suited for logistics and distribution functions, largely due to its strong transport connections and strategic accessibility. At the same time, the district includes nearby residential neighborhoods that are increasingly affected by the expansion of industrial activity.

Existing logistics and industrial zones often occupy valuable open space, reduce access to greenery, and disrupt visual connections within the urban environment. This creates a significant urban conflict: while logistics development offers economic opportunity and infrastructural efficiency, it also introduces noise, heavy traffic, and environmental pressures that can negatively impact the quality of life for residents. As a result, Prague 9 represents both an opportunity and a constraint for future logistics growth, requiring careful urban planning strategies that balance industrial productivity with housing, public space, and environmental quality.

- Proposed Site
- Open Landscape
- Agriculture / Fields
- Productive Agriculture
- Residential / Mixed
- Public Facilities
- Historic / Protected
- Commercial / Industrial

The site is located within the center of Prague 9 at a critical urban junction, acting as a transitional zone between the district’s industrial heritage and its growing residential fabric. To the east and north, the site is surrounded by high-density residential clusters and mixed-use urban zones, while its southern edge connects directly to a significant ecological corridor composed of active recreation areas and public green space.

Due to its strategic proximity to major transportation infrastructure and arterial roads, the area presents a strong opportunity for a logistics hub capable of supporting both the surrounding neighborhoods and the wider urban network.

However, the conventional “big-box” logistics typology would be unsuitable in this context. Such developments typically operate as isolated industrial enclaves, creating physical and visual barriers that disconnect surrounding communities while consuming valuable urban land.

On a site of this scale and importance, a purely industrial solution would prevent the brownfield from being reintegrated into the city and would fail to address the growing need for accessible public space and ecological continuity within Prague 9.

Instead of functioning as a closed logistical object, the site has the potential to become a hybrid urban landscape — one that combines logistics infrastructure with public greenery, recreation, and community-oriented spaces.

This project aims to transform logistics from a segregated industrial activity into an integrated urban system, allowing the development to support economic functionality while simultaneously contributing to environmental quality, public life, and the long-term revitalization of the district.



- | | | |
|---|--|--|
| ■ Residential | ■ Industrial | ■ Parks & Sports |
| ■ Public Utility Areas | ■ Proposed Com. & Log. | ■ Cultural Monument |



The Venn diagram below illustrates the overlapping relationships and activities present within the site and its surrounding urban context.

The site is shaped by three primary spatial conditions: the residential community, the abandoned industrial district, and the brownfield landscape with the potential to become a public park. Together, these elements create an opportunity for a more integrated and socially connected urban environment.

The residential zones represent the community-oriented aspect of the site, prioritizing quality of life through accessible public areas, local events, and spaces that encourage outdoor activity and social engagement. These environments aim to strengthen neighborhood interaction and reduce the isolation often associated with dense urban living.



Although each layer of the site serves different functions, they intersect through a shared central idea: social interaction. The project therefore proposes a hybrid urban condition where logistics, public space, and residential life are not separated, but interconnected through accessible landscapes, community-oriented programs, and shared public experiences.

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The abandoned industrial district represents the commercial and infrastructural layer of the site. Existing industrial buildings and large open plots provide the necessary framework for a logistics hub while also offering opportunities for adaptive reuse through cafés, shops, and mixed commercial functions. Rather than remaining isolated industrial objects, these spaces could be redesigned to engage directly with the surrounding community and support everyday public life.

The brownfield area introduces the possibility of a large public park that acts as a transitional landscape between logistics infrastructure and residential neighborhoods. As a green buffer, it could provide ecological value, recreation along the riverbank, playgrounds, pedestrian connections, and spaces for informal gathering. In doing so, the park becomes both environmental infrastructure and social infrastructure.

The axonometric view of the topography illustrates how existing urban features relate to one another across the site. It highlights different patterns of movement, including pedestrians walking through the green spaces, logistical flows such as truck routes, and activity along the green corridor. It also emphasizes the connection between existing residential green spaces and the future park currently in development.



 New Park Proposal

 Ecological Corridor

- Logistics Hub
- Public Flow

Existing Residential Space

Logistical Truck Flow

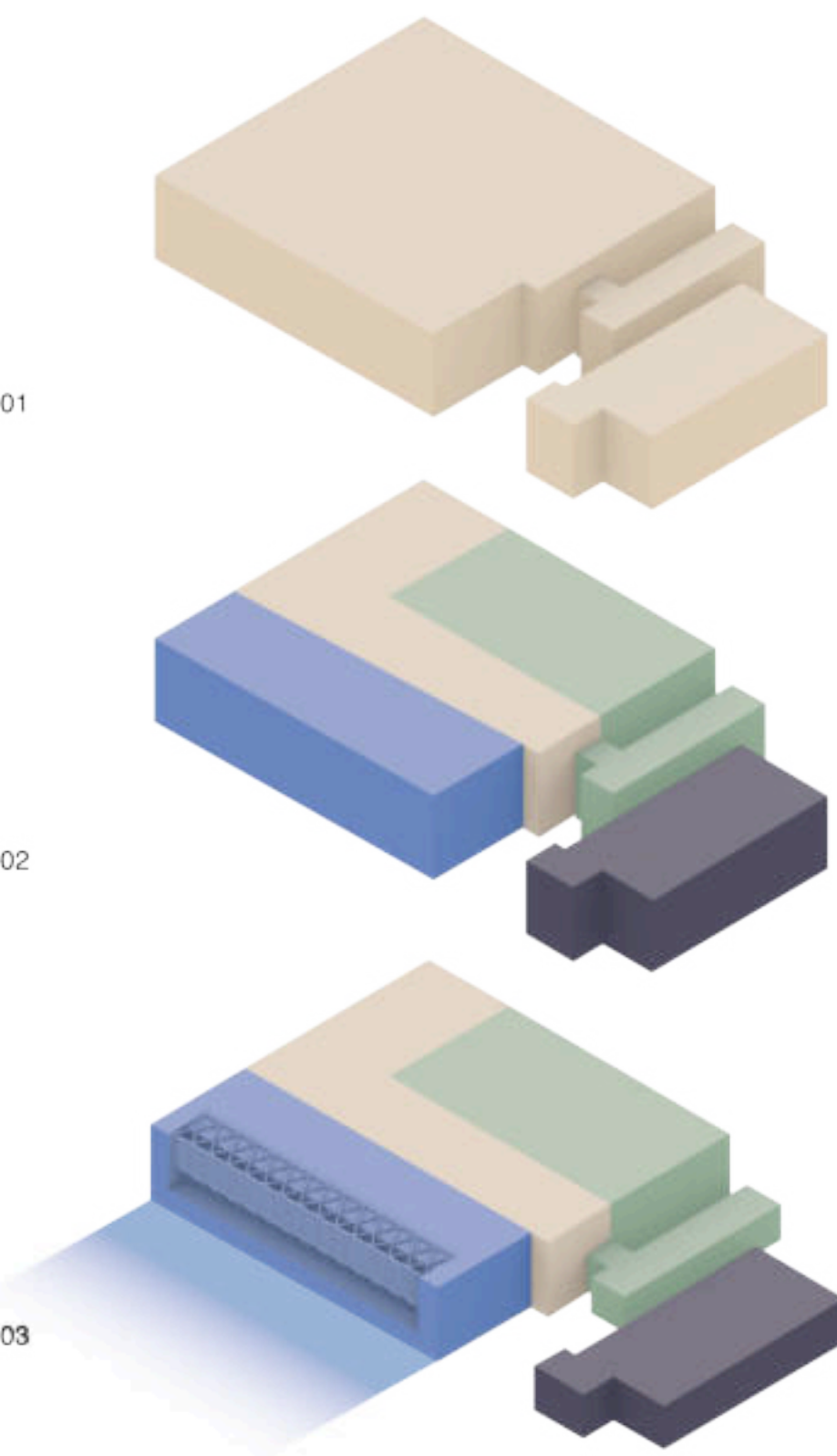
ARCHIP

AD2 - LOGISTICS

concept

This diagram illustrates the conceptual evolution of the project, beginning with the existing building and progressing through a series of transformations toward the simplified final form.

Each stage highlights how the massing and spatial organization are refined throughout the design process. The colors represent the different activities accommodated within the building, indicating the proportion of space dedicated to each function and clarifying the relationship between program distribution and overall form.

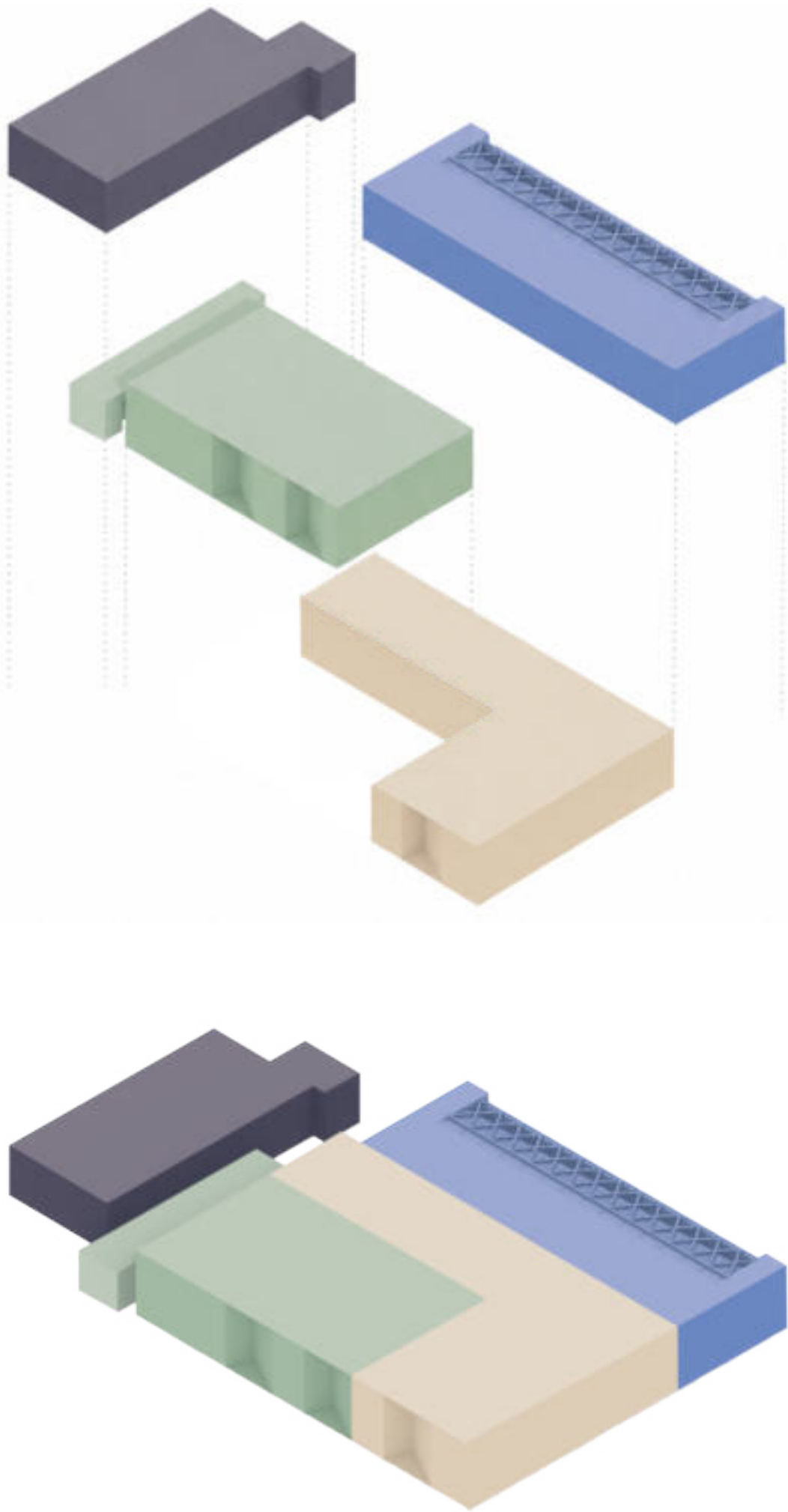


The initial industrial plant consists of three separate buildings, each serving a different function within the manufacturing process. The main building is primarily accessed from the east side, providing the principal point of entry and circulation. However, the structure lacks substantial entrances on the west side, limiting accessibility and restricting connections to the surrounding site and adjacent buildings.

The next step involves organizing the interior into distinct functional zones based on operational needs and site context. The west side is transformed into a publicly accessible commercial area, creating a stronger connection with the surrounding urban environment. The northern section remains dedicated to manufacturing, preserving the industrial character and existing production activities. Meanwhile, the south-east portion is adapted into loading docks and large-scale storage facilities, improving logistical efficiency and service access. Through this redistribution of functions, the building evolves into a multi-purpose complex that responds more effectively to both industrial requirements and the conditions of the site.

The final step focuses on redesigning the commercial section to introduce natural light from the adjacent park, indicated in green. By opening the structure toward this landscape, the building gains a more welcoming and transparent character, strengthening the relationship between the interior spaces and the public realm. This intervention transforms the former industrial envelope into a more accessible and inviting environment, encouraging public access directly from the park. At the same time, the building establishes itself as an architectural landmark through the introduction of an expressive exo-skeleton beam system, which structurally supports the two commercial floors while defining the project's visual identity.

- Logistics
- Storage
- Commercial
- Manufacturing



The first KPI focuses on carbon emissions of a reconstruction and adaptive reuse of an existing building.

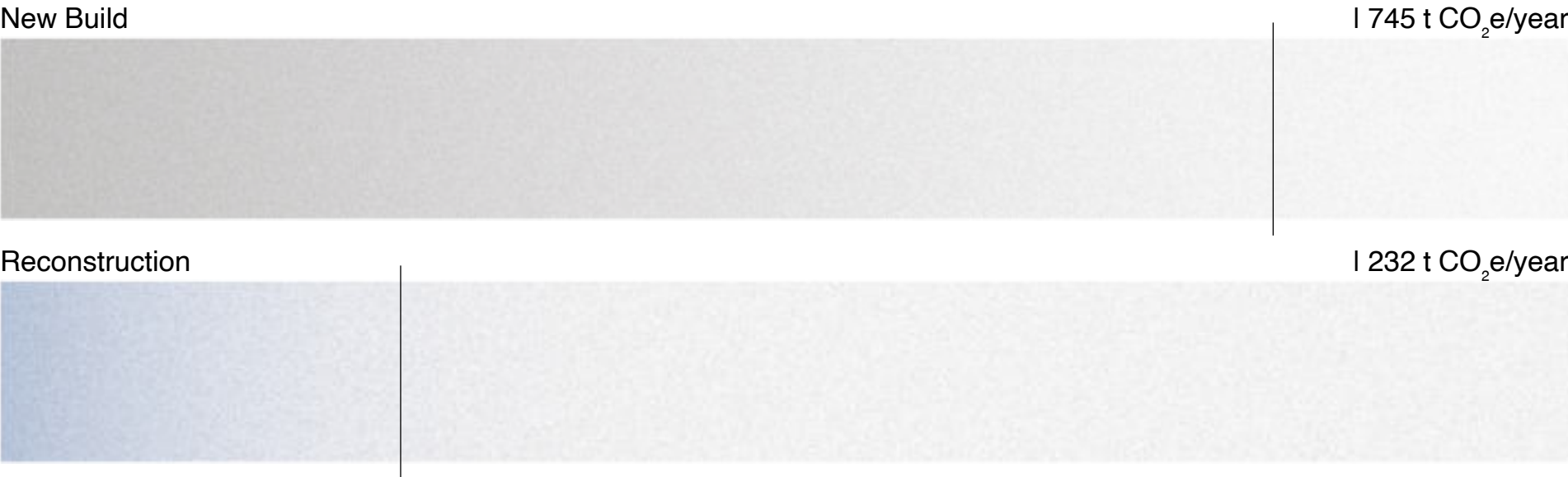
Embodied carbon refers to the greenhouse gas emissions associated with the extraction, manufacturing, transportation, and construction of building materials throughout the building process. In most conventional construction projects, the largest contributors to embodied carbon are the structural frame, foundations, and primary envelope systems due to the intensive use of materials such as concrete, steel, and glass.

By reusing the existing structure and foundations, the project avoids many of the most environmentally damaging stages of construction. Large quantities of demolition waste are prevented, while the need for new raw materials is significantly reduced.

This not only lowers carbon emissions but also decreases energy consumption associated with manufacturing and transporting replacement materials.

Retaining the existing structural system also reduces disruption to the surrounding site and shortens the overall construction timeline. The reconstruction strategy is estimated to reduce embodied carbon by approximately 70% when compared to a fully new-build alternative of similar scale and function.

This represents a major environmental benefit and aligns strongly with contemporary sustainable design principles, which increasingly prioritise reuse and adaptation over demolition and replacement.



Operational LCA Information

Warehouse size: 6,400 m²

Electricity consumption: 30,000 kWh/month

Estimated forklifts: 6 electric forklifts

Estimated pallets handled: 160,000 pallets/year

In addition to the environmental advantages, reconstruction also provides significant economic benefits.

Reusing major structural components lowers material and construction costs, reduces labour requirements associated with extensive demolition and excavation, and minimises the need for expensive new foundation works.

These financial savings create opportunities for investment in other areas of building performance and occupant comfort. In this project, a portion of the saved resources can be redirected towards improving the thermal performance of the building envelope through enhanced insulation systems and adaptive façade technologies.

Operational Carbon

Annual Electricity Consumption

30,000 (avg. monthly) x 12 = 360,000 kWh/year

Czech GEF - 0.45 kg CO₂/kWhI Total: 360 mWh/year

Pallets Handled

Approximately 25 pallets/m²/year

6400 x 25 = 160,000 pallets/yearI Total: 160k pallets/year

Operational Carbon Intensity

162,000 / 160,000 = 1.01 kg CO₂/pallet

I Total: 1.01 kg CO₂/pallet

Building Operational Emissions

E = 360,000 x 0.45 = 162,000

162,000 kg CO₂/year = 162 t CO₂/yearI Total: 162 t CO₂/year

Forklift Estimate

6 Forklifts (electric)

6 x 8000 = 48,000 kWh/yearI Total: 48 mWh / year

Embodied Carbon

Embodied Carbon

Retrofit: 330kg CO₂e/m

6400 x 330 = 2,112,000 kg CO₂eI Total: 2,122 t CO₂e

Lifespan

Estimated 30 years lifespan

2112 / 30 = 70.4I Total: 70.4 t CO₂e/year

Annualized LCA

162 + 70.4 = 232.4 t CO₂e/year

I Total: 232.4 t CO₂e/year

Carbon Intensity Per Pallet

232,400 / 160,000 = 1.45 kg CO₂/pallet

I Total: 1.45kg CO₂/pallet

The second KPI is energy efficiency, which is a critical factor in the long-term performance and sustainability of the building during its operational life.

Reducing energy consumption not only lowers operational and maintenance costs, but also significantly decreases the building’s environmental impact by reducing carbon emissions and overall resource demand.

To improve performance, the building envelope will incorporate enhanced thermal and acoustic insulation, helping to minimise heat transfer and maintain stable indoor temperatures throughout the year. In addition, the façade will feature adaptive panels that respond dynamically to the sun’s movement, optimising shading and daylight levels while reducing unwanted solar heat gain.

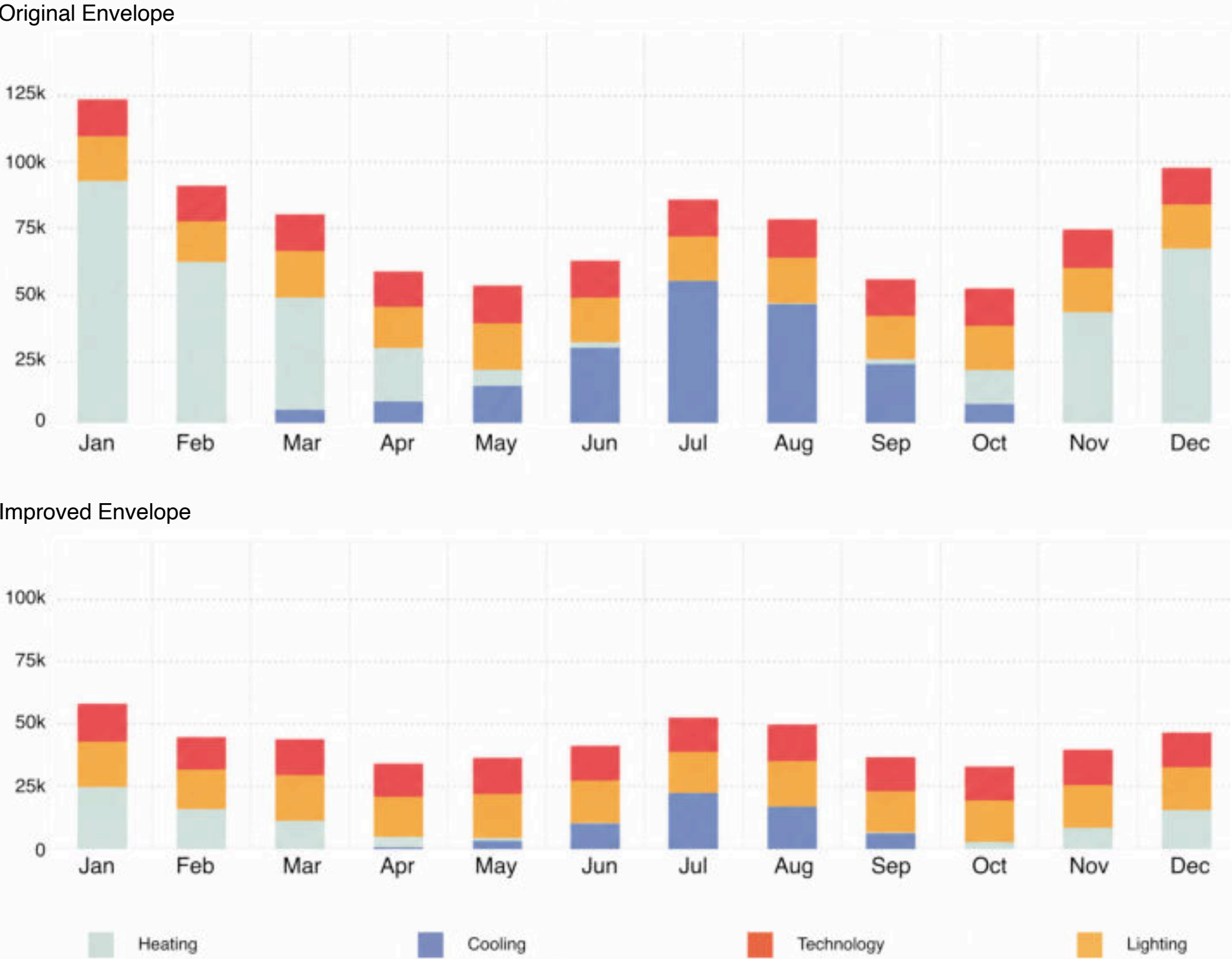


As a result of these combined strategies, the building is expected to achieve an average annual reduction of approximately 37.5% in kilowatt-hour energy consumption compared to a conventional building of similar scale and function.

This substantial decrease demonstrates the effectiveness of passive design measures and smart façade technologies in creating a more energy-efficient, cost-effective, and environmentally responsible building.

The graphs below illustrate the monthly energy consumption of the building before and after the renovation of the envelope. Since lower energy consumption directly corresponds to lower carbon emissions during the building’s operation, the comparison clearly demonstrates the positive environmental impact of the renovation measures. A substantial improvement can be observed across summer / winter months, with the post-renovation data showing a consistent reduction in energy demand throughout the year.

This decrease indicates that the upgraded building envelope significantly improved the thermal performance of the structure by reducing heat losses during colder months and minimizing unwanted heat gains during warmer periods.

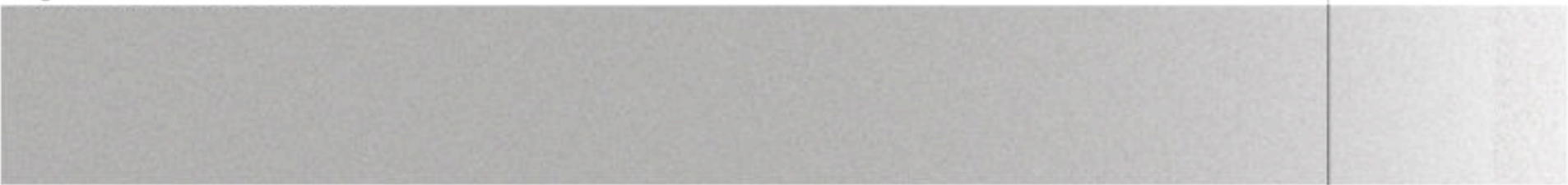


The third KPI focuses on the safety and wellbeing of residents during logistics operations. In the previous site layout, residents were required to cross several roads in order to access neighbouring residential areas. While this arrangement may have been acceptable under conditions of low traffic volume and limited vehicle size, the introduction of a logistics facility at the end of the road significantly increases the presence of heavy vehicles and associated safety risks.

To address this issue, a dedicated pedestrian underpass will be introduced, allowing residents to move safely between residential zones without directly interacting with vehicular traffic.

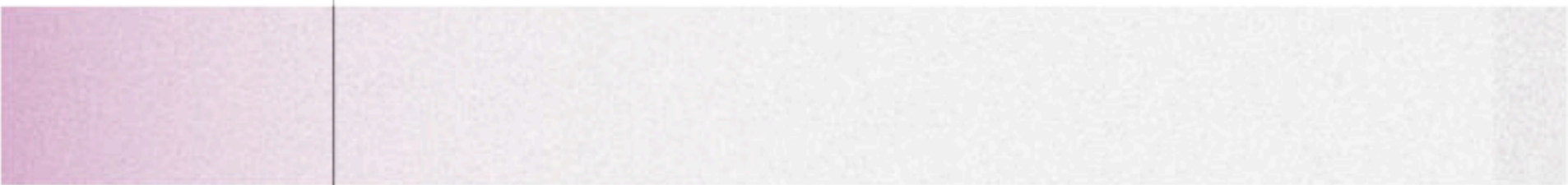
This intervention greatly reduces the likelihood of conflicts between pedestrians and logistics vehicles, particularly large trucks operating throughout the site.

Original Traffic Intersections



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New Traffic Intersections

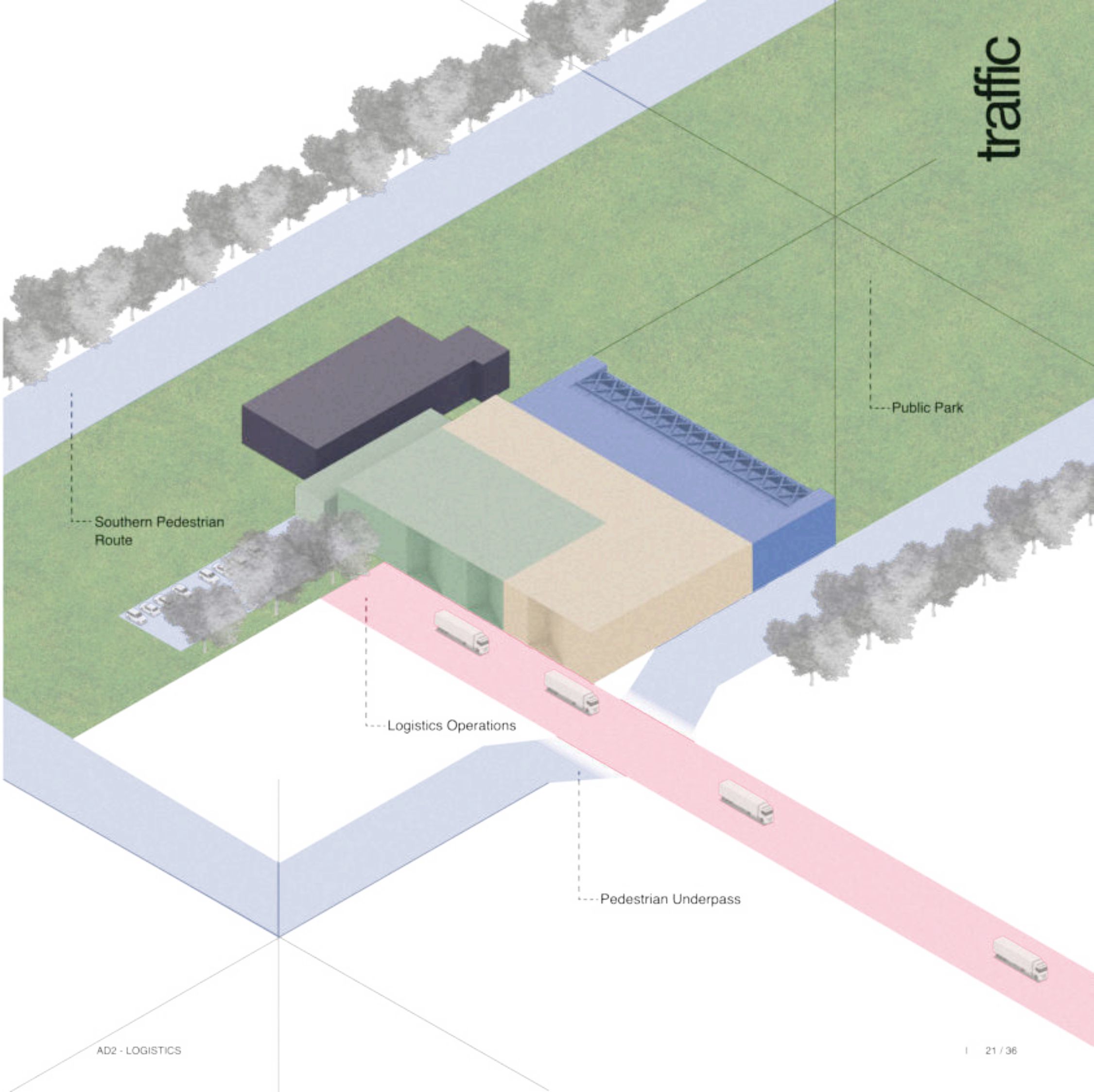


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Accessibility has also been considered within the design, with wheelchair users and individuals with reduced mobility able to safely navigate the area through the southern pedestrian route.

By minimising the interaction between residents and heavy logistics traffic, the overall level of pedestrian safety is expected to improve substantially. In addition to reducing physical risk, the underpass establishes a clear and visible separation between residential and industrial functions.

This spatial distinction helps ensure that logistics activities remain contained within designated operational areas, preventing unnecessary overlap with residential movement and creating a safer, more comfortable environment for the community.



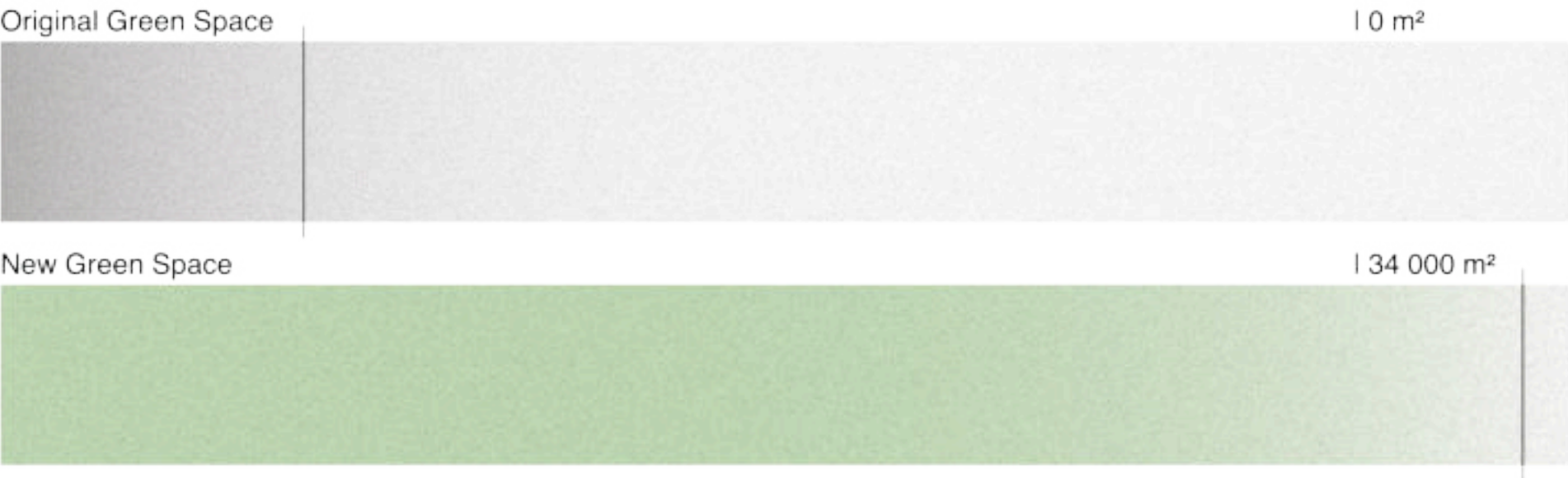
The fourth KPI focuses on improving biodiversity within the newly introduced green spaces surrounding the development. As part of the proposal, a new public park will be created on the western side of the commercial area. Rather than functioning solely as a recreational space for residents, the park is designed to support a wider urban ecosystem by accommodating local birds, insects, and other small city wildlife alongside human activity.

This approach recognises the importance of biodiversity in creating healthier and more sustainable urban environments. Urban green spaces often prioritise aesthetics and recreation while offering limited ecological value. In contrast, this proposal aims to create a more balanced landscape that supports both social and environmental functions. By incorporating a variety of planting types, vegetation densities, and habitat conditions, the park can encourage a broader range of species to inhabit and move through the area.

Increasing biodiversity within the site also contributes to improved air quality, urban cooling, stormwater absorption, and overall environmental resilience.

To achieve this, the park is divided into four distinct landscape zones, each designed to serve a specific purpose while contributing to the overall ecological strategy of the site.

By dividing the park into these specialised zones, the project creates a layered and multifunctional landscape that balances recreation, accessibility, and ecological performance. The strategy not only improves the quality of life for residents but also strengthens biodiversity within the urban environment, demonstrating how landscape design can actively contribute to both community wellbeing and environmental sustainability.



Zone A

Consists of planted areas surrounding the commercial spaces, primarily featuring potted plants and managed greenery. These areas help soften the urban environment, improve visual quality, and create a more welcoming atmosphere for pedestrians and visitors. In addition, flowering plants within this zone can provide food sources for pollinating insects such as bees and butterflies.

Zone B

Made up of short-cut grass areas intended for general public use and recreation. These open spaces provide flexibility for activities such as walking, sitting, and informal gatherings, while maintaining clear sightlines and accessibility throughout the park. Although more controlled in appearance, these spaces still contribute to the overall green coverage of the site.

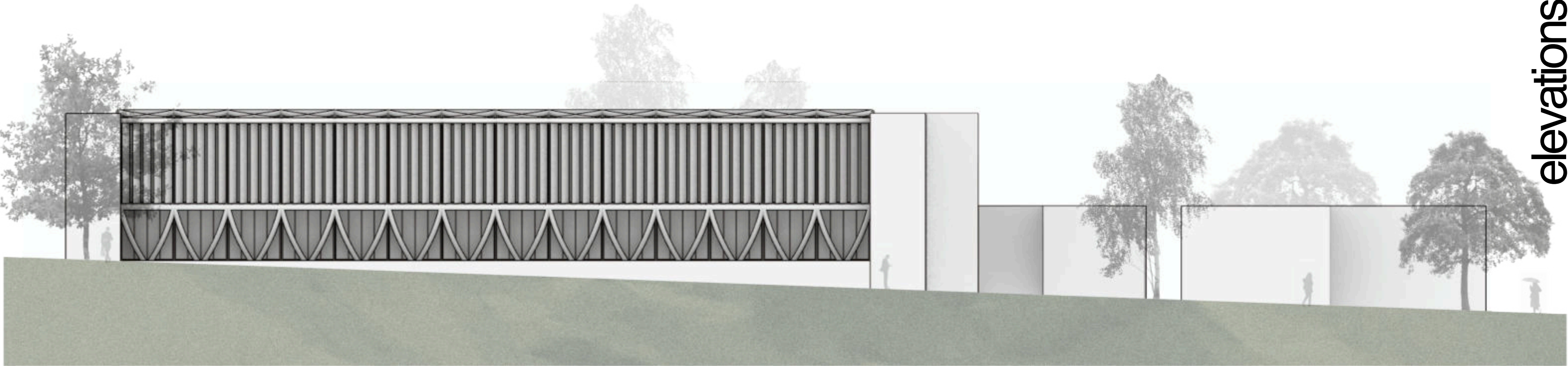
Zone C

Introduces taller grass and bushier vegetation specifically intended as a more natural recreational environment for dogs and their owners. The increased vegetation density creates a softer transition between highly maintained public areas and more natural habitats, while also offering shelter and habitat opportunities for insects and small animals.

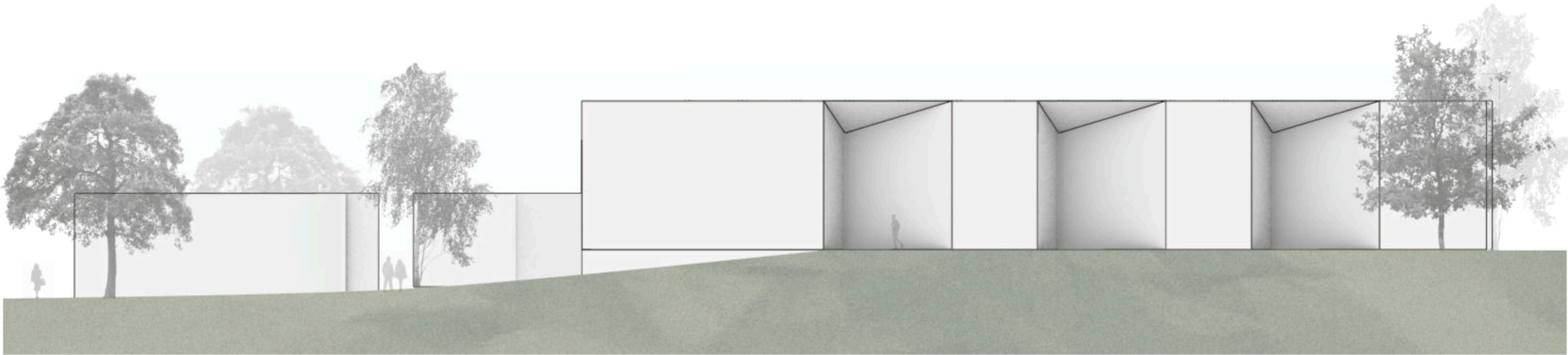
Zone D

Forms the most ecologically focused section of the park, consisting of lush greenery, uncut grass, and large trees designed to support local wildlife. This area acts as a small urban habitat where birds, insects, and other city animals can find food, shelter, and nesting opportunities with minimal human disturbance. The inclusion of mature trees and dense vegetation also improves shade, reduces urban heat island effects, and enhances the long-term ecological value of the site. By dividing the park into these specialised zones, the project creates a layered and multifunctional landscape that balances recreation, accessibility, and ecological performance. The strategy not only improves the quality of life for residents but also strengthens biodiversity within the urban environment, demonstrating how landscape design can actively contribute to both community wellbeing and environmental sustainability.

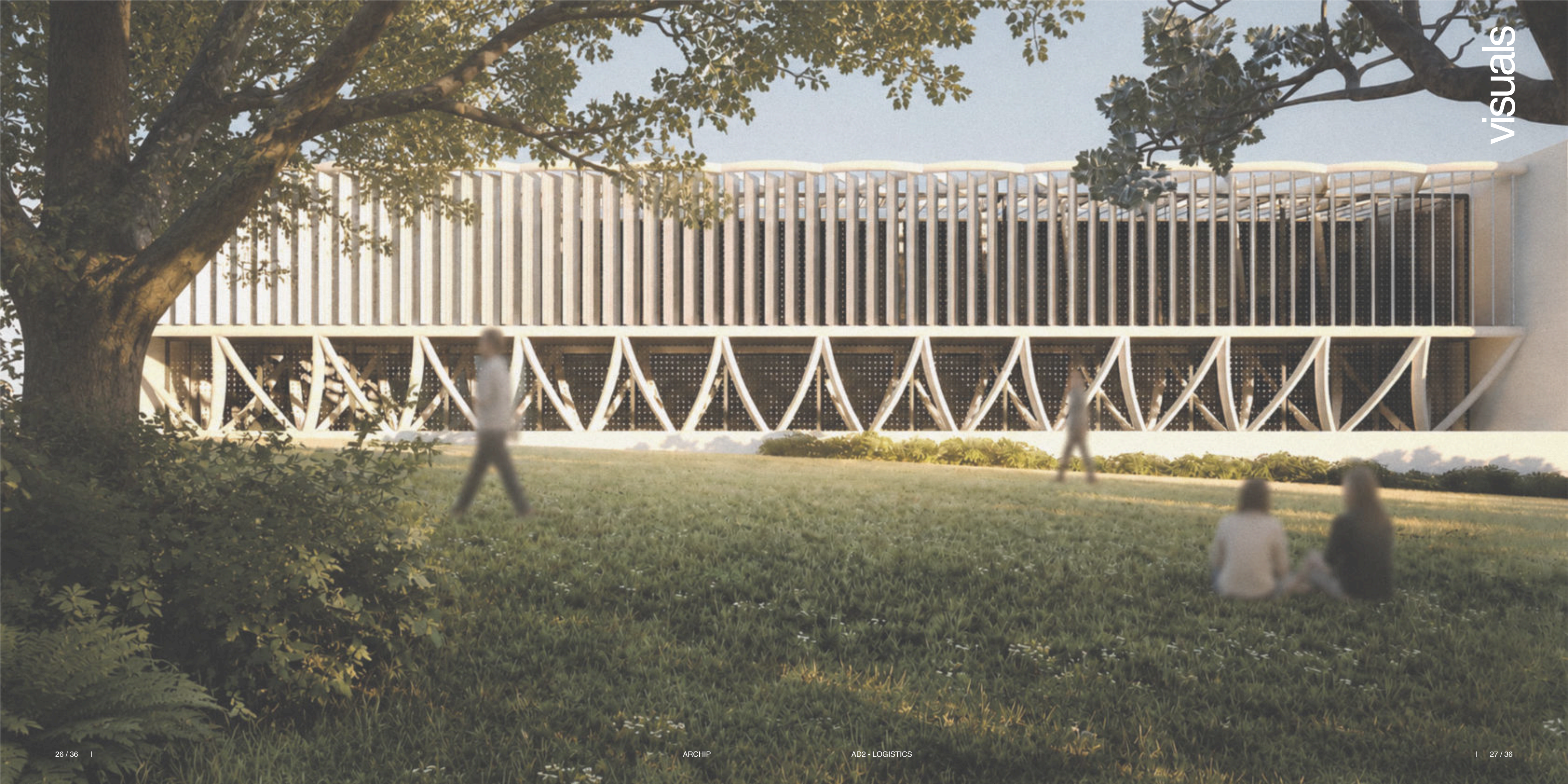




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